

Dynamical calibration of a metallicity-dependent mass–luminosity relation from *Gaia* wide binaries

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ABSTRACT

Context. The mass–luminosity relation (MLR) of low-mass main-sequence stars links observed luminosities to stellar masses throughout Galactic and exoplanet astronomy, yet its metallicity dependence is calibrated almost entirely by stellar-evolution models. Wide binaries resolved by *Gaia* carry dynamical information on the sum of their component masses through their projected relative velocities, and their components share a common birth composition.

Aims. We aim to measure, directly from the orbital motions of *Gaia* wide binaries, how the main-sequence MLR in the *Gaia* *G* band depends on metallicity, and to quantify the departures of the inferred relation from the PARSEC models.

Methods. We analyse 14 876 main-sequence–main-sequence wide binaries within 1 kpc. A modular Bayesian model first combines per-star metallicities inferred from *Gaia* XP spectra with the PARSEC colour–magnitude relation to obtain a posterior for the shared metallicity of each system. A second module models the stellar mass as a hard-monotone tensor-product B-spline surface in absolute magnitude and metallicity, expressed relative to the PARSEC prediction, and confronts it with the mass-scaled projected-velocity statistic $u = \Delta v_{\perp} \sqrt{r_{\perp}}$ through a Rice-convolved likelihood with an outlier component. The shape of the orbital-velocity distribution is sampled jointly with the MLR in four independent metallicity bins.

Results. The inferred MLR depends significantly on metallicity. Relative to PARSEC, stars in the most metal-rich bin ($[M/H] = 0.3\text{--}0.6$) are 8–14% more massive at $5 \leq M_G \leq 13$, while those in the most metal-poor bin ($[M/H] = -1.0\text{--}-0.5$) are 3–6% less massive for $M_G \leq 11$ and $\approx 20\%$ less massive at $M_G = 13$. Near-solar bins agree with PARSEC to within $\approx 7\%$ for $M_G \leq 9$ but fall 10–18% below it at $M_G = 13$. The metallicity sensitivity $d \ln M / d[M/H]$ rises from 0.27 ± 0.04 at $M_G = 5$ to 0.75 ± 0.09 at $M_G = 13$, 1.4–1.8 times the PARSEC value. When the velocity-distribution shape is instead fixed to its orbital-population prediction, the inferred masses exceed PARSEC by $\approx 30\%$ at $M_G \approx 8$; the absolute mass scale is therefore controlled by the assumed velocity distribution.

Conclusions. Within the adopted family of velocity distributions, wide-binary dynamics favour a metallicity dependence of the main-sequence MLR with the sign predicted by stellar models, metal-rich stars being more massive at fixed M_G , but with a larger amplitude than PARSEC. Both the absolute mass scale and the metallicity sensitivity are degenerate with the intrinsic velocity distribution of wide binaries, whose posterior lies at the edge of the explored range; an independent constraint on that distribution is the main requirement for a model-independent dynamical calibration.

Key words. binaries: visual – stars: fundamental parameters – stars: low-mass – stars: abundances – methods: statistical – astrometry

1. Introduction

The mass of a main-sequence star is the quantity that most directly sets its luminosity and evolution, but it can be measured only for stars whose orbits can be characterised. Empirical mass–luminosity relations (MLRs) of low-mass stars are therefore anchored to a few hundred eclipsing and astrometric binaries with precise dynamical masses (Delfosse et al. 2000; Torres et al. 2010; Mann et al. 2019). These samples are dominated by stars of near-solar composition. Mann et al. (2019) found no significant metallicity dependence of the M_{K_s} –mass relation, where models predict the dependence to be weak; in optical bands such as *Gaia* *G* the predicted dependence is much stronger, and it is in practice supplied by stellar-evolution models such as PARSEC (Bressan et al. 2012). At fixed absolute magnitude, a metal-rich star is expected to be more massive than a metal-poor one, because higher opacity lowers the luminosity at fixed mass. The amplitude of this effect, which reaches tens of per cent across the metallicity range of the Galactic disc, has not been measured dynamically for large samples.

Wide binaries offer a statistical route to such a measurement. With projected separations of $10^2\text{--}10^4$ AU, their orbital periods greatly exceed the *Gaia* baseline, so individual orbits cannot be solved. The projected relative velocity $\Delta v_{\perp}$ and projected separation $r_{\perp}$ of a bound pair are, however, related to the total mass by Kepler’s third law. For a population with a known distribution of eccentricities, phases, and orientations, the combination $u = \Delta v_{\perp} \sqrt{r_{\perp}}$ scales as $\sqrt{M_{\text{tot}}}$ times a dimensionless variable whose distribution is fixed by orbital geometry. Hwang et al. (2024) used this statistic to derive dynamical masses across the Hertzsprung–Russell diagram from the catalogue of El-Badry et al. (2021). The same statistic underlies tests of gravity in the low-acceleration regime with wide binaries (Chae 2023; Banik et al. 2024), which rely on an assumed MLR to predict the Newtonian velocity scale.

The components of a wide binary formed together and share their birth composition. Metallicities of the individual stars can now be inferred for millions of *Gaia* sources from the low-resolution BP/RP (XP) spectra (Gaia Collaboration et al. 2023; Li et al. 2026). Each binary thus supplies two noisy estimates of a single metallicity and one dynamical constraint on the sum of

two masses. Combined over a large sample, these data constrain the MLR as a function of both luminosity and metallicity.

Turning this into a measurement requires care at three points. First, the XP metallicities have magnitude-dependent systematics and heavy-tailed errors, which must be calibrated before they are used to assign binaries to metallicities. Second, the mass enters the likelihood only through the scale of a broad velocity distribution, so the result depends on the assumed shape of that distribution and on the treatment of contaminants such as chance alignments and hierarchical triples. Third, a flexible MLR must remain physically ordered: brighter stars must be more massive at fixed composition.

In this paper we present a modular Bayesian framework that addresses these points and apply it to 14 876 main-sequence wide binaries within 1 kpc. A first module infers a posterior for the shared metallicity of each system from the component XP metallicities and the PARSEC colour–magnitude relation. A second module represents the MLR as a hard-monotone B-spline correction to the PARSEC mass surface and fits it to the wide-binary velocities while marginalising over each system’s metallicity posterior. We describe the sample in Sect. 2, the model in Sect. 3, and the results in Sect. 4. Section 5 discusses the degeneracy between the mass scale and the velocity distribution and the limitations of the present analysis, and Sect. 6 summarises our conclusions.

2. Data

2.1. Wide-binary sample

The parent sample is the *Gaia* EDR3 wide-binary catalogue of [El-Badry et al. \(2021\)](#), restricted to pairs classified as two main-sequence stars (MSMS). We retain systems with parallax $\varpi_1 > 1$ mas (distance below 1 kpc), projected separation $r_\perp < 10^4$ AU, a proper-motion difference measured at more than 3σ , and a low chance-alignment probability as defined in that catalogue (all selected systems have $\mathcal{R} \leq 0.07$; [state the adopted threshold](#)). All components have astrometric fidelities ≥ 0.69 from the classifier of [Rybizki et al. \(2022\)](#) ([state the adopted threshold](#)). Each component was classified as a single star by a model comparison of single-star and unresolved-binary fits to its XP spectrum ([Li et al., in prep.](#); [confirm reference and the \$\Delta\text{BIC}\$ threshold](#)), which removes the most luminous unresolved companions that would bias the photometric mass. Extinctions were taken from the 3D dust map of [Edenhofer et al. \(2024\)](#) ([confirm](#)); the selected systems have $A_G < 0.1$ mag. The dereddened colours $C_{jk} = (G_{\text{BP}} - G_{\text{RP}})_{0,jk}$ and absolute magnitudes $x_{jk} = M_{G,jk}$ follow from the *Gaia* photometry ([Riello et al. 2021](#)), these extinctions, and the parallax of the [primary](#). Systems with a non-finite or non-positive velocity statistic, metallicity uncertainty, or colour uncertainty are removed. The final sample contains 14 876 binaries (29 752 stars) with $2.6 \lesssim M_G \lesssim 13.9$ and median projected separation 1 900 AU.

2.2. Velocity statistic

For each system we form the mass-scaled velocity statistic of [Hwang et al. \(2024\)](#),

$$u_j = \Delta v_{\perp,j} \sqrt{r_{\perp,j}} = 4.74 \frac{\Delta \mu_j}{\varpi_j} \sqrt{r_{\perp,j}/\text{AU}} \text{ km s}^{-1} \text{ AU}^{1/2}, \quad (1)$$

where $\Delta \mu_j$ is the magnitude of the proper-motion difference in mas yr^{−1} and ϖ_j is the parallax in mas. Its uncertainty is $\sigma_{u,j} =$

$u_j/(\Delta \mu/\sigma_{\Delta \mu})_j$, with $(\Delta \mu/\sigma_{\Delta \mu})_j$ the significance of the proper-motion difference. The sample has median $u = 22 \text{ km s}^{-1} \text{ AU}^{1/2}$ and median $\sigma_u = 2.5 \text{ km s}^{-1} \text{ AU}^{1/2}$.

2.3. XP metallicities

Per-star metallicities $\widehat{z}_{jk}$ and their reported uncertainties $\sigma_{z,jk}$ are inferred from the *Gaia* XP spectra with the J-CaPS forward model ([Li et al. 2026](#)). We use the raw per-star values, not any product that already enforces equal metallicity within a binary; such a product would count the equal-composition assumption of our likelihood twice. The median reported uncertainty is 0.05 dex, and the raw values span $-1.0 \leq \widehat{z} \leq 0.58$ with 90% of stars between -0.57 and $+0.20$.

3. Method

3.1. Overview and notation

We infer the stellar mass M as a function of absolute magnitude $x \equiv M_G$ and metallicity $Z \equiv [\text{M}/\text{H}]$. Binary systems are indexed by j , their components by $k \in \{1, 2\}$, and the nodes of a metallicity grid by q . Hats denote observed quantities; $\log_{10}$ is used for stellar-mass logarithms and $\log$ for natural logarithms.

The analysis is modular ([Liu et al. 2009](#); [Plummer 2015](#)) (Fig. 1). Stage 1a calibrates the relative, magnitude-dependent behaviour of the XP metallicities from within-binary differences. Stage 1b combines the calibrated metallicities with the PARSEC colour–magnitude relation and a population distribution to obtain, for each system, a posterior probability mass $\bar{P}_{jq}$ for its shared metallicity on a fixed grid $\{Z_q\}$. Stage 2 predicts the two component masses at every grid node, evaluates the dynamical likelihood of u_j , and sums over nodes with weights $\bar{P}_{jq}$. Information flows in one direction only: the dynamics do not recalibrate the metallicity zero point, the colour scatter, or the population distribution.

3.2. Stage 1: system metallicities

3.2.1. XP measurement model

Each star carries an independent Student- t measurement of the shared system metallicity Z_j ,

$$\widehat{z}_{jk} | Z_j \sim \mathcal{T}_\nu \left[Z_j + \delta_z + b_G(x_{jk}), (\sigma_{z,jk}^2 + s_z^2)^{1/2} \right] \equiv \mathcal{L}_{jk}^{\text{XP}}(Z_j), \quad (2)$$

where the second and third arguments are the location and scale. The function $b_G(x)$ is a magnitude-dependent relative bias, s_z an additional per-star scatter, ν the degrees of freedom, and δ_z a global zero point. In Stage 1a, b_G , s_z , and ν are estimated by maximum likelihood from the within-binary differences $\widehat{z}_{j1} - \widehat{z}_{j2}$, which do not depend on Z_j or δ_z ; they are then held fixed.

3.2.2. Colour–magnitude likelihood

The zero point δ_z is not constrained by metallicity differences. It is set in Stage 1b by requiring the observed colours to agree with the PARSEC colour $C_P(x, Z)$ at the inferred metallicity. For a component with formal colour uncertainty $\sigma_{C,jk} = (2.5/\ln 10)(\text{SNR}_{\text{BP}}^{-2} + \text{SNR}_{\text{RP}}^{-2})^{1/2}$,

$$C_{jk} | Z_j \sim \mathcal{T}_4 \left[C_P(x_{jk}, Z_j), (\sigma_{C,jk}^2 + s_C^2)^{1/2} \right] \equiv \mathcal{L}_{jk}^{\text{CMD}}(Z_j), \quad (3)$$

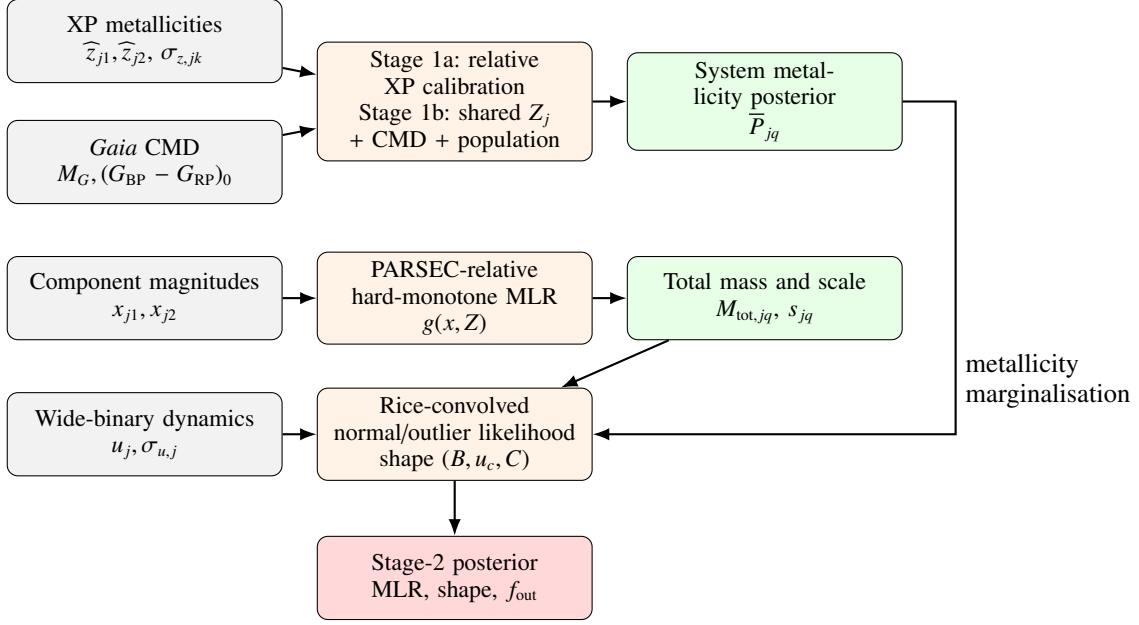


Fig. 1. Structure of the inference. Stage 1 returns a posterior probability mass over the shared metallicity of every system. Stage 2 propagates this uncertainty into the dynamical fit of the MLR but does not feed back into Stage 1.

with an excess scatter $s_C \sim \text{Half-Normal}(0.1 \text{ mag})$. Because PARSEC colours of cool dwarfs are less reliable, only components whose PARSEC effective temperature exceeds 4000 K at every grid metallicity enter this term:

$$m_{jk} = \mathbb{I} \left[\min_q T_{\text{eff,P}}(x_{jk}, Z_q) \geq 4000 \text{ K} \right]. \quad (4)$$

The mask is independent of the proposed metallicity, so the set of stars that enter the CMD term does not change during sampling. It retains 12 167 stars; 8 247 systems have at least one and 3 920 have two such components. The PARSEC reference surface is the average over isochrones with $9.3 < \log_{10}(\text{age/yr}) < 10.0$, so age is not a latent variable.

3.2.3. System posteriors

The population distribution $p_{\text{pop}}(Z | \omega)$ is a six-component mixture with Dirichlet(1, ..., 1) weights ω , and $\delta_z \sim \mathcal{N}(0, 0.3 \text{ dex})$. On an 81-node grid spanning $-1.0 \leq Z \leq 0.6$, the posterior probability mass for system j is

$$\bar{P}_{jq} \propto w_q \left\langle p_{\text{pop}}(Z_q | \omega) \prod_{k=1}^2 \mathcal{L}_{jk}^{\text{XP}}(Z_q) \left[\mathcal{L}_{jk}^{\text{CMD}}(Z_q) \right]^{m_{jk}} \right\rangle_{\psi}, \quad (5)$$

where w_q are trapezoidal weights, $\sum_q \bar{P}_{jq} = 1$, and the average is over the Stage-1b posterior of $\psi = (\delta_z, s_C, \omega)$. The global parameters are constrained by the product of the per-system evidences. Stage 1b was sampled with the No-U-Turn Sampler (NUTS; Hoffman & Gelman 2014) (chains and draws of the formal run).

3.3. Stage 2: a PARSEC-relative hard-monotone MLR

3.3.1. Spline representation

Let $g_P(x, Z) = \log_{10}[M_P(x, Z)/M_{\odot}]$ be the PARSEC mass surface. The fitted surface is a tensor-product cubic B-spline

(de Boor 2001),

$$g(x, Z) = \sum_{i=1}^{K_x} \sum_{h=1}^{K_Z} \Theta_{ih} B_i(x) C_h(Z) = g_P^*(x, Z) + \Delta_*(x, Z), \quad (6)$$

where $g_P^* = \sum \Pi_{ih} B_i C_h$ is the least-squares projection of g_P onto the same basis and $\Delta_* = \sum D_{ih} B_i C_h$ with $D = \Theta - \Pi$ is the fitted correction. The projection error $g_P^* - g_P$ has an rms of 0.002 dex and a maximum of 0.019 dex. We use $K_x \times K_Z = 8 \times 4$ basis functions with clamped knot vectors $\kappa_x = (3.5^{(4)}, 5.5, 7.5, 9.5, 11.5, 13.5^{(4)}) \text{ mag}$ and $\kappa_Z = (-1.0^{(4)}, 0.6^{(4)}) \text{ dex}$, where superscripts give multiplicities. The metallicity dependence is therefore a single cubic polynomial segment.

3.3.2. Hard monotonicity

At fixed metallicity a fainter star must not be more massive, and at fixed magnitude we require that a more metal-rich star is not less massive. We impose

$$\Theta_{i+1,h} \leq \Theta_{ih}, \quad \Theta_{i,h+1} \geq \Theta_{ih}, \quad (7)$$

which, because the derivatives of a B-spline are non-negative combinations of adjacent coefficient differences, guarantees $\partial g / \partial x \leq 0$ and $\partial g / \partial Z \geq 0$ everywhere in the domain. The constraints act on the total surface Θ , not on the correction D , which may take either sign. Appendix A gives the constructive parametrisation.

3.3.3. Regularisation and solar anchor

The correction is shrunk towards PARSEC and towards smoothness through

$$\log \pi(D) = -\frac{1}{2\tau_D^2} \sum_{i,h} D_{ih}^2 - \frac{1}{2} \sum \left(\frac{D_{ih}^{(2,x)}}{\tau_x} \right)^2 - \frac{1}{2} \sum \left(\frac{D_{ih}^{(2,Z)}}{\tau_Z} \right)^2 - \frac{1}{2} \sum \left(\frac{D_{ih}^{(xZ)}}{\tau_{xZ}} \right)^2 + \text{const}, \quad (8)$$

where $D^{(2,x)}$ and $D^{(2,Z)}$ are second differences and $D^{(x,Z)}$ the mixed difference of the coefficients, normalised by the Greville-abscissa spacing so that they retain units of dex (Eilers & Marx 1996). We adopt $\tau_D = 0.10$ dex and $\tau_x = \tau_Z = \tau_{xZ} = 0.05$ dex. The absolute mass scale is anchored softly at the Sun,

$$g(4.67, 0) \sim \mathcal{N}(0, [0.01 / \ln 10]^2), \quad (9)$$

i.e. a 1% constraint on the mass of a solar-luminosity, solar-metallicity star.

3.4. Dynamical likelihood

3.4.1. Normal and outlier components

For a bound binary on a Keplerian orbit, the projected relative velocity and separation combine to $u = \sqrt{M_{\text{tot}}/M_{\odot}} \tilde{u}$, where $\tilde{u}$ depends only on eccentricity, orbital phase, and orientation. Marginalising these over an assumed orbital population gives a distribution $p(\tilde{u})$ that is common to all systems. We parametrise it with the functional form of Hwang et al. (2024),

$$p_N(\tilde{u} | B, u_c, C) \propto \tilde{u} \exp\left[-B\tilde{u}^2 - \exp\left(\frac{\tilde{u} - u_c}{C}\right)\right], \quad 0 < \tilde{u} \leq 80, \quad (10)$$

normalised on its support. The reference values $(B, u_c, C) = (2.544 \times 10^{-3}, 35.67, 3.1)$ were obtained by fitting this form to simulated $\tilde{u}$ for an orbital population (specify eccentricity distribution and separation range used in the simulation).

Pairs that are not simple bound binaries, such as hierarchical triples with an unresolved inner companion or residual chance alignments, produce velocities that do not scale with the photometric mass. We describe them with an outlier component defined directly on the latent observed-frame amplitude v ,

$$p_O(v) = \frac{\phi[(v - 40)/13]}{13 [2\Phi(40/13) - 1]} \mathbb{I}(0 \leq v \leq 80), \quad (11)$$

a normal distribution with mean 40 and dispersion $13 \text{ km s}^{-1} \text{ AU}^{1/2}$ truncated to $[0, 80]$, independent of the mass and metallicity of the system.

3.4.2. Measurement model

The measured u_j is the modulus of a two-dimensional velocity difference with approximately isotropic Gaussian errors, so its conditional density given the true amplitude v is the Rice distribution (Rice 1945),

$$R(u | v, \sigma) = \frac{u}{\sigma^2} \exp\left[-\frac{u^2 + v^2}{2\sigma^2}\right] I_0\left(\frac{uv}{\sigma^2}\right). \quad (12)$$

For a mass scale $s = \sqrt{M_{\text{tot}}/M_{\odot}}$ the two component likelihoods are

$$L_{N,j}(s) = \int_0^{80s} R(u_j | v, \sigma_{u,j}) p_N\left(\frac{v}{s}\right) \frac{dv}{s}, \quad (13)$$

$$L_{O,j} = \int_0^{80} R(u_j | v, \sigma_{u,j}) p_O(v) dv, \quad (14)$$

and the mixture is $L_j(s) = (1 - f_{\text{out}})L_{N,j}(s) + f_{\text{out}}L_{O,j}$ with $f_{\text{out}} \sim \text{Beta}(3, 12)$.

3.4.3. Metallicity marginalisation

At metallicity node Z_q the component masses and mass scale are $M_{jkq} = M_{\odot} 10^{g(x_{jk}, Z_q)}$ and $s_{jq} = [(M_{j1q} + M_{j2q})/M_{\odot}]^{1/2}$. The Stage-2 likelihood of system j is

$$\mathcal{L}_j^{\text{dyn}} = \sum_{q=1}^{81} \bar{P}_{jq} L_j(s_{jq}), \quad (15)$$

which propagates the metallicity uncertainty of each system into its mass. Neither the population distribution nor the quadrature weights are applied a second time, because both are contained in $\bar{P}_{jq}$.

3.5. Metallicity bins and velocity-distribution shape

The shape parameters (B, u_c, C) strongly affect the inferred mass scale (Sect. 4.1), and they need not be the same for binaries of different metallicity, since the wide-binary fraction and separation distribution depend on metallicity (Hwang et al. 2021). We therefore fit four metallicity bins independently, each with its own MLR surface, shape parameters, and outlier fraction. Systems are assigned by the median of their metallicity posterior to the bins $[-1.0, -0.5]$, $[-0.5, 0.0]$, $[0.0, 0.3]$, and $[0.3, 0.6]$, which contain 186, 9 618, 4 687, and 385 systems. Within a bin every system retains its full 81-node posterior in Eq. (15). The MLR of each bin is reported at the bin midpoint.

The shape parameters receive uniform priors in $\log B$, $\log u_c$, and $\log C$ within the box $B \in [0.7, 3.3] \times 10^{-3}$, $u_c \in [24, 44]$, and $C \in [2.5, 13.5]$. To make sampling tractable, Eq. (13) is pre-computed for every system on a logarithmic grid of mass scales $s \in [0.25, 2.5]$ at $3 \times 3 \times 3$ nodes spanning this box. At a general point, the normal-component likelihood is the probability mixture of the 27 node likelihoods with trilinear weights in the logarithmic shape coordinates. This mixture is itself a normalised distribution in u , although it is not identical to Eq. (10) evaluated at the interpolated parameters.

3.6. Priors and posterior sampling

The MLR is parametrised by 35 unconstrained coordinates (Appendix A), to which the three shape parameters are added (Table 1). Each bin was sampled with NUTS (Hoffman & Gelman 2014) using four chains of 1 500 warm-up and 2 000 retained draws. Each bin was sampled twice with independent seeds and jittered initial values. All Stage-2 runs completed without divergent transitions. Convergence diagnostics are listed in Table 5.

4. Results

4.1. Fixed velocity-distribution shape

We first fit a single MLR surface to all 14 876 systems with the shape of the normal component fixed at its orbital-population reference, $(B, u_c, C) = (2.544 \times 10^{-3}, 35.67, 3.1)$, and only f_{out} free (Fig. 2). The fit is well converged (maximum rank-normalised $\bar{R} = 1.003$, minimum bulk and tail effective sample sizes $> 4 700$, no divergences). The inferred MLR is not a small correction to PARSEC: at $M_G \approx 8$ the dynamical masses exceed PARSEC by $\approx 30\%$ at all metallicities, for example $0.901^{+0.018}_{-0.017}$ versus $0.696 M_{\odot}$ at $M_G = 7$ and $[\text{M}/\text{H}] = 0$. The excess is confined to $5 \lesssim M_G \lesssim 11$; it vanishes at the solar anchor and changes sign for the faintest stars. The outlier fraction is $f_{\text{out}} = 0.207 \pm 0.006$.

Table 1. Stage-2 parameters and priors.

Parameter	Prior
$c_0 = \Theta_{11}$	$\mathcal{N}(\Pi_{11}, 0.10)$ dex
a_h, b_i, r_{ih}	$\mathcal{N}(0, 1)$
$\log \lambda_x$	$\mathcal{N}(\log 0.186, 0.7)$
$\log \lambda_z$	$\mathcal{N}(\log 0.131, 0.7)$
$\log B$	$\mathcal{U}(\log 0.0007, \log 0.0033)$
$\log u_c$	$\mathcal{U}(\log 24, \log 44)$
$\log C$	$\mathcal{U}(\log 2.5, \log 13.5)$
f_{out}	$\text{Beta}(3, 12)$
$\tau_D; \tau_x, \tau_z, \tau_{xz}$	0.10; 0.05 dex (fixed)
Solar anchor	$g(4.67, 0) = 0 \pm 0.0043$ dex

Notes. λ_x and λ_z set the scale of the monotone increments (Appendix A); their reference values are the mean PARSEC coefficient increments.

Figure 2 also shows a variant in which the outlier component is defined on $\tilde{u}$, that is, it scales with the photometric mass like the normal component. This variant raises the mass excess to ≈ 38 –47% and lowers f_{out} to 0.149 ± 0.005 : when the outlier distribution scales with the mass, the outlier term no longer penalises a larger mass scale, and high-velocity systems are absorbed by the normal component. Defining the outlier component in the observed frame, Eq. (11), removes this coupling, and we adopt it throughout. The remaining $\approx 30\%$ excess is large compared with the few-per-cent precision of empirical MLRs of nearby M dwarfs (Mann et al. 2019), and it has the shape of a mass-scale distortion that peaks where the sample is densest. This indicates that the fixed $p_N(\tilde{u})$ does not describe the observed velocity distribution, and that the MLR absorbs the mismatch.

4.2. Joint MLR and velocity-distribution fits in metallicity bins

Figure 3 and Table 2 present the MLR when the shape parameters are sampled jointly with the MLR in four independent metallicity bins. Freeing the shape removes most of the mass excess of Sect. 4.1: in every bin the inferred masses lie within $\approx 15\%$ of PARSEC for $M_G \leq 11$. The two independent runs per bin agree to better than 0.25% in the median mass at every magnitude.

The residuals relative to PARSEC depend systematically on metallicity. In the most metal-rich bin ($[M/H] = 0.3$ –0.6, evaluated at +0.45) the stars are more massive than PARSEC by 8–14% over $5 \leq M_G \leq 13$, and the excess differs from zero by more than 1σ everywhere in this range. In the most metal-poor bin (-1.0 to -0.5 , at -0.75) the masses lie 3–6% below PARSEC for $M_G \leq 11$ and 20% below at $M_G = 13$. The two near-solar bins, which contain 96% of the sample, agree with PARSEC to within $\approx 7\%$ for $M_G \leq 9$ and fall below it by 10–18% at $M_G = 13$. In the $[0, 0.3)$ bin the residual changes from +7% at $M_G = 7$ to -13% at $M_G = 11$, a stronger magnitude dependence than in any other bin.

4.3. Metallicity dependence of the MLR

Because the four bins are fitted independently, the difference between the two outer bins provides a dynamical measurement of the metallicity sensitivity of the MLR. Between $[M/H] = -0.75$ and +0.45 we find $d \ln M / d[M/H] = 0.27 \pm 0.04$ at $M_G = 5$, 0.31 ± 0.07 at $M_G = 7$, 0.41 ± 0.08 at $M_G = 9$, 0.58 ± 0.08

at $M_G = 11$, and 0.75 ± 0.09 at $M_G = 13$ (Table 3). The sign agrees with stellar-evolution theory, and the sensitivity increases towards fainter stars, as in PARSEC. The amplitude, however, is 1.4–1.8 times the PARSEC value at every magnitude. Across the 1.2 dex between the outer bins, the mass at fixed M_G thus increases by 37% at $M_G = 5$ and by a factor of 2.4 at $M_G = 13$, compared with 20% and a factor of 1.8 in PARSEC.

4.4. Velocity-distribution shape and outlier fraction

Table 4 lists the posteriors of the shape parameters and of f_{out} . All bins prefer a smaller B and u_c than the orbital-population reference: $u_c \approx 28$ –30 in the three bins with $[M/H] > -0.5$, compared with 35.67, and B within 35% of the lower edge of the prior box. In the $[-0.5, 0)$ bin, which holds 65% of the sample, $B = 0.716^{+0.025}_{-0.012} \times 10^{-3}$ and $C = 2.69^{+0.28}_{-0.14}$ lie against the lower limits $B = 0.7 \times 10^{-3}$ and $C = 2.5$ of the explored range. A smaller u_c moves the exponential cut-off of Eq. (10) to lower $\tilde{u}$, so that a given observed u requires a larger $\sqrt{M_{\text{tot}}}$; this is the direction that allows the MLR to return towards PARSEC. The metal-poor bin is the exception, with $u_c = 39.6^{+2.8}_{-3.4}$, above the reference value.

The outlier fraction is 0.17–0.23 in the three bins with $[M/H] < 0.3$ and $0.107^{+0.025}_{-0.022}$ in the metal-rich bin.

4.5. Sampling diagnostics

Table 5 summarises the convergence of the eight Stage-2 runs. Six runs have $\hat{R} < 1.003$ and minimum effective sample sizes above 1900. The two runs of the $[0.0, 0.3)$ bin mix more slowly (minimum ESS 213 and 137, maximum $\hat{R} = 1.007$ and 1.026); their medians agree to 0.25%, but their intervals, which are broad and asymmetric in Tables 2 and 4, should be regarded as provisional.

5. Discussion

5.1. Mass scale and velocity distribution

The dynamical likelihood constrains the total mass only through the ratio of the observed velocity scale to the intrinsic scale of $p_N(\tilde{u})$. A uniform rescaling $\tilde{u} \rightarrow \alpha \tilde{u}$ of the intrinsic distribution is exactly compensated by $M_{\text{tot}} \rightarrow M_{\text{tot}}/\alpha^2$. Without external information, the absolute mass scale and the velocity distribution are therefore degenerate; in our model the degeneracy is broken only by the solar anchor (Eq. 9), the PARSEC-relative shrinkage (Eq. 8), and the functional form of Eq. (10). The comparison between Sects. 4.1 and 4.2 shows this directly: changing the shape from the orbital-population reference to the posterior shape moves the MLR by up to 30% at $M_G \approx 8$.

The same degeneracy acts on the metallicity sensitivity of Sect. 4.3. Each bin has its own shape parameters, so a change of the mass scale in one bin can be compensated by its shape independently of the other bins. The fitted shapes move in the corresponding direction: the metal-poor bin, which lies below PARSEC, has $u_c = 39.6$, whereas the metal-rich bin, which lies above PARSEC, has $u_c = 27.8$ (Table 4). Under uniform priors on the logarithmic shape parameters, the data prefer this combination, and the PARSEC-relative shrinkage of Eq. (8) pulls each bin towards PARSEC, so that the regularisation biases the measured sensitivity towards the PARSEC value rather than away from it. A genuine metallicity dependence of the intrinsic velocity distribution, for example through the separation or eccentric-

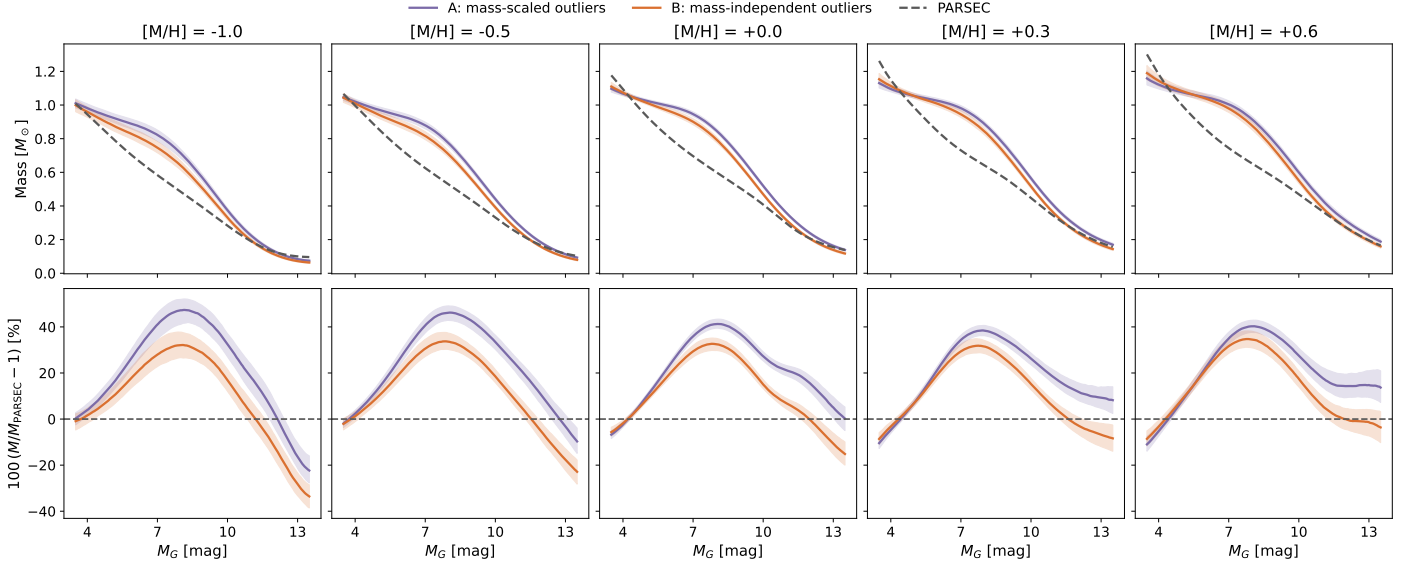


Fig. 2. MLR inferred with the velocity-distribution shape fixed to its orbital-population reference, for the full sample. Top: mass as a function of M_G at five metallicities; bottom: relative difference from PARSEC. Orange: outlier component defined on the observed u (Eq. 11); purple: outlier component that scales with the photometric mass. Dashed lines show PARSEC; bands are 16–84% posterior intervals.

Table 2. Dynamical mass relative to PARSEC, $100 (M/M_p - 1)$, in per cent.

Bin	N	$[M/H]_{\text{eval}}$	$M_G = 5$	7	9	11	13
$[-1.0, -0.5)$	186	-0.75	$-5.6^{+3.7}_{-3.9}$	$-3.3^{+5.8}_{-5.6}$	$-2.5^{+6.4}_{-6.2}$	$-5.9^{+5.9}_{-5.7}$	$-19.5^{+6.0}_{-5.6}$
$[-0.5, 0.0)$	9 618	-0.25	$+0.7^{+1.7}_{-1.6}$	$+3.7^{+3.4}_{-3.3}$	$+5.9^{+3.4}_{-3.1}$	$-1.0^{+3.0}_{-3.2}$	$-9.9^{+5.1}_{-4.8}$
$[0.0, 0.3)$	4 687	+0.15	$+4.5^{+1.5}_{-1.6}$	$+7.2^{+5.2}_{-7.0}$	$-2.0^{+5.7}_{-10.9}$	$-12.7^{+5.1}_{-9.2}$	$-17.8^{+5.9}_{-6.3}$
$[0.3, 0.6]$	385	+0.45	$+7.9^{+3.1}_{-2.6}$	$+13.9^{+5.9}_{-5.8}$	$+13.9^{+7.7}_{-7.0}$	$+13.0^{+7.5}_{-7.4}$	$+9.1^{+8.3}_{-7.6}$

Notes. Posterior medians with 16th and 84th percentiles from the first run of each bin, evaluated at the bin midpoint $[M/H]_{\text{eval}}$. The corresponding PARSEC masses at $M_G = 7$ are 0.600, 0.657, 0.714, and 0.741 $M_\odot$.

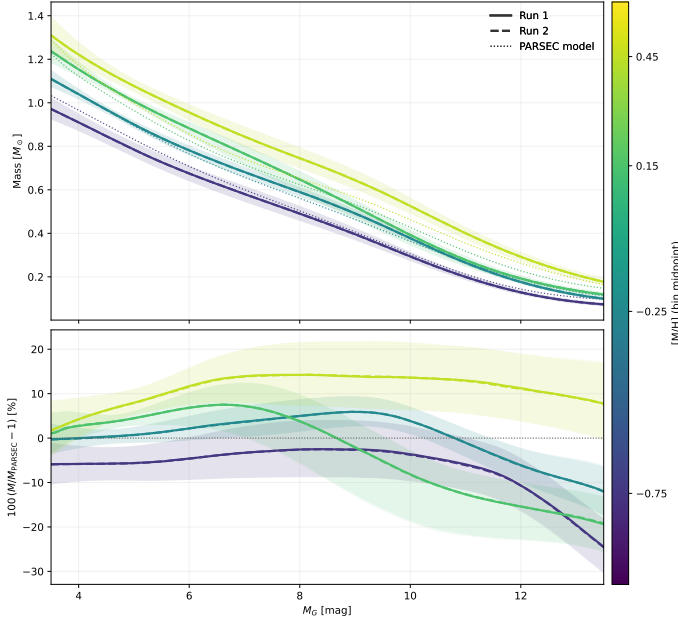


Fig. 3. MLR from the joint fits of the MLR and velocity-distribution shape in four independent metallicity bins, evaluated at the bin midpoints (colour). Top: posterior median mass (solid and dashed: two independent runs) and PARSEC (dotted). Bottom: relative difference from PARSEC. Bands are pointwise 16–84% intervals.

Table 3. Metallicity sensitivity of the MLR between the outer bins.

M_G	$d \ln M / d[M/H]$	PARSEC	Ratio
5	$0.265^{+0.041}_{-0.039}$	0.152	1.74
7	$0.311^{+0.067}_{-0.065}$	0.176	1.77
9	$0.406^{+0.076}_{-0.074}$	0.276	1.47
11	$0.580^{+0.075}_{-0.074}$	0.428	1.36
13	$0.746^{+0.086}_{-0.085}$	0.493	1.52

Notes. Finite difference of $\ln M$ between $[M/H] = +0.45$ and -0.75 , divided by 1.2 dex, from independent posterior draws of the two bins.

ity distributions (Hwang et al. 2021, 2022), would nevertheless be traded against the MLR of each bin, in either direction. The measured $d \ln M / d[M/H]$ is therefore the metallicity sensitivity under the assumption that the shape family of Eq. (10) describes each bin; it is not independent of that assumption.

5.2. Why the orbital-population shape is disfavoured

The posterior shapes of the near-solar bins have a lower cut-off u_c and a broader Gaussian core (smaller B) than the orbital-population reference. Several effects can produce such a difference. The reference assumes an eccentricity distribution and a

Table 4. Velocity-distribution shape and outlier fraction.

Bin	$10^3 B$	u_c	C	f_{out}
$[-1.0, -0.5)$	$0.90^{+0.26}_{-0.14}$	$39.6^{+2.8}_{-3.4}$	$5.2^{+4.0}_{-2.0}$	$0.173^{+0.051}_{-0.044}$
$[-0.5, 0.0)$	$0.716^{+0.025}_{-0.012}$	$29.3^{+0.4}_{-0.4}$	$2.69^{+0.28}_{-0.14}$	$0.230^{+0.007}_{-0.007}$
$[0.0, 0.3)$	$0.74^{+0.08}_{-0.03}$	$29.1^{+4.2}_{-0.8}$	$3.8^{+3.7}_{-0.9}$	$0.188^{+0.014}_{-0.038}$
$[0.3, 0.6]$	$0.93^{+0.39}_{-0.18}$	$27.8^{+1.9}_{-1.6}$	$3.6^{+1.8}_{-0.8}$	$0.107^{+0.025}_{-0.022}$
Reference	2.544	35.67	3.1	–
Prior range	0.7–3.3	24–44	2.5–13.5	Beta(3, 12)

Notes. Posterior medians and 16–84% intervals from the first run of each bin.

Table 5. Convergence diagnostics of the Stage-2 runs.

Bin	Run 1		Run 2	
	max $\hat{R}$	min ESS	max $\hat{R}$	min ESS
$[-1.0, -0.5)$	1.0020	2620	1.0014	3139
$[-0.5, 0.0)$	1.0024	2079	1.0009	1922
$[0.0, 0.3)$	1.0067	213	1.0255	137
$[0.3, 0.6]$	1.0015	2247	1.0021	2258

Notes. Four chains of 2 000 retained draws per run; no divergent transitions in any run.

separation range that need not match the selected sample; wide binaries in *Gaia* have a separation-dependent eccentricity distribution (Hwang et al. 2022). Projection effects and the selection on the significance of $\Delta\mu$ remove low-velocity systems, which reshapes the observed distribution relative to the intrinsic one. Finally, unresolved companions in hierarchical triples raise u beyond the Keplerian value for the photometric mass; if their distribution in u differs from the truncated normal of Eq. (11), they distort the fitted normal component. The lower f_{out} of the metal-rich bin is consistent with a metallicity-dependent triple or contamination fraction, but f_{out} is a mixture weight between two fixed shapes and should not be read as a physical triple fraction.

5.3. Comparison with PARSEC

The sign of the metallicity dependence agrees with PARSEC, and at the metallicity of most of the sample the dynamical MLR agrees with PARSEC to within $\approx 7\%$ for $M_G \lesssim 9$. Two differences stand out. First, the dynamical metallicity sensitivity is 1.4–1.8 times the PARSEC value. Second, the faintest stars ($M_G \approx 13$, $M \approx 0.1\text{--}0.2 M_\odot$) are 10–20% less massive than PARSEC at $[M/H] < 0.3$. This regime contains few systems (93 systems contribute to the neighbourhood of $M_G = 13$ in the full sample), so the MLR there is constrained largely by the spline regularisation and the monotonicity of brighter stars, and PARSEC colours and luminosities of fully convective stars carry larger systematic uncertainties. The faint-end deficit should therefore be regarded as tentative. A direct comparison with empirical MLRs of nearby M dwarfs (Mann et al. 2019), which requires a transformation between the K_s and G bands, is deferred to future work.

For the wide-binary tests of gravity (Chae 2023; Banik et al. 2024), which predict Newtonian velocities from photometric masses, a metallicity dependence of the MLR that exceeds the

model prediction implies that a metallicity-independent MLR misestimates $\sqrt{M_{\text{tot}}}$ by several per cent for the metal-poor and metal-rich tails of a sample.

5.4. Limitations

The results are conditional on the modelling choices listed below.

- *Shape prior range.* In the two most populous bins the posterior of B , and in the $[-0.5, 0)$ bin also that of C , lies against the lower edge of the explored range. The 27-node likelihood table cannot represent shapes outside this range, so the reported MLR for these bins is the best fit within the range rather than an unconstrained optimum, and the true posterior may extend to smaller B . Extending the range, or replacing the table by direct quadrature of Eq. (13), is required before the absolute masses of these bins can be quoted without qualification.
- *Shape interpolation.* The normal-component likelihood at a general shape is a mixture of node likelihoods, not Eq. (10) at the interpolated parameters. Its accuracy relative to direct quadrature, and the sensitivity of the fits to single-precision arithmetic, have not been tested for the full bins.
- *Initialisation.* The two runs per bin were intended to start from different shape configurations, but the initial values were passed to the sampler in unconstrained coordinates without the required transformation. The runs therefore test reproducibility under different seeds rather than robustness to widely separated starting points.
- *Mixing in the $[0.0, 0.3)$ bin.* The two runs in this bin have minimum ESS of 213 and 137 and $\hat{R}$ up to 1.026; their intervals are provisional.
- *Stage-1 calibration.* The Stage-1b posterior used here contains three divergent transitions. The Stage-1a parameters b_G , s_z , and ν are held at point estimates, and the cross-system covariance induced by the shared Stage-1b parameters is not propagated.
- *Modular cut.* The dynamics cannot update the metallicity calibration, and the Stage-2 uncertainties do not include uncertainty in M_G , extinction, age, or the PARSEC model family.
- *Metallicity bins.* Systems are assigned to bins by their posterior median, so systems near a boundary contribute to the bin they are assigned to while retaining their full metallicity posterior. Each bin constrains a full $(M_G, [M/H])$ surface, which we evaluate only at the bin midpoint; the metal-poor and metal-rich bins contain 186 and 385 systems.
- *PARSEC dependence.* The MLR is a regularised correction to PARSEC, anchored at the solar point. Residuals relative to PARSEC are therefore not independent measurements of PARSEC errors; they are shrunk towards zero where the data are weak.

6. Conclusions

We have measured the metallicity dependence of the main-sequence mass–luminosity relation in the *Gaia* G band from the relative motions of 14 876 wide binaries within 1 kpc. A modular Bayesian model infers the shared metallicity of each binary from its two XP metallicities and the PARSEC colour–magnitude relation, and propagates this posterior into a hard-monotone, PARSEC-relative spline MLR constrained by a Rice-convolved likelihood of the mass-scaled velocity statistic. Our main conclusions are the following.

1. With the velocity-distribution shape fixed to its orbital-population reference, the inferred masses exceed PARSEC by $\approx 30\%$ near $M_G \approx 8$ at all metallicities. The absolute dynamical mass scale is controlled by the assumed intrinsic velocity distribution.
2. When the shape is sampled jointly with the MLR in four metallicity bins, the MLR agrees with PARSEC to within $\approx 15\%$ for $M_G \leq 11$ in all bins and to within $\approx 7\%$ for $M_G \leq 9$ in the near-solar bins, which contain 96% of the sample.
3. The MLR depends significantly on metallicity. Relative to PARSEC, stars with $[M/H] = 0.3\text{--}0.6$ are 8–14% more massive over $5 \leq M_G \leq 13$, and stars with $[M/H] = -1.0\text{--}0.5$ are 3–6% less massive for $M_G \leq 11$.
4. The metallicity sensitivity $d \ln M / d[M/H]$ rises from 0.27 ± 0.04 at $M_G = 5$ to 0.75 ± 0.09 at $M_G = 13$. It has the sign predicted by PARSEC but is 1.4–1.8 times larger.
5. The preferred velocity distribution of the near-solar bins has a lower cut-off and a broader core than the orbital-population reference and lies at the edge of the explored parameter range.

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An absolute dynamical calibration of the MLR requires an independent constraint on the intrinsic velocity distribution of wide binaries, for example from forward modelling of the separation-dependent eccentricity distribution and of the *Gaia* selection, together with a treatment of hierarchical triples. The same constraint would turn the metallicity sensitivity measured here, which is currently conditional on the adopted shape family, into a model-independent measurement, and the longer astrometric baseline of *Gaia* DR4 will reduce the velocity uncertainties that limit the metal-poor and metal-rich bins.

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Appendix A: Constructive monotone reparametrisation

The total coefficient grid Θ is generated from unconstrained raw variables rather than by rejecting proposals that violate inequalities. Define $\text{softplus}(t) = \log(1 + e^t)$ and $\text{sigmoid}(t) = (1 + e^{-t})^{-1}$. With positive scales $\lambda_x = e^{\log \lambda_x}$ and $\lambda_Z = e^{\log \lambda_Z}$, construct the first row and column as

$$\Theta_{11} = c_0, \quad (\text{A.1})$$

$$\Theta_{1h} = \Theta_{1,h-1} + \lambda_Z \text{softplus}(a_h), \quad h = 2, \dots, K_Z, \quad (\text{A.2})$$

$$\Theta_{i1} = \Theta_{i-1,1} - \lambda_x \text{softplus}(b_i), \quad i = 2, \dots, K_x. \quad (\text{A.3})$$

The first row therefore increases with Z , whereas the first column decreases with x . Interior points are visited in increasing $i + h$. For each point, define

$$L_{ih} = \Theta_{i,h-1}, \quad U_{ih} = \Theta_{i-1,h}, \quad (\text{A.4})$$

and set

$$\Theta_{ih} = L_{ih} + (U_{ih} - L_{ih}) \text{sigmoid}(r_{ih}), \quad i, h \geq 2. \quad (\text{A.5})$$

If all previously constructed points obey the two monotone orders, then

$$L_{ih} = \Theta_{i,h-1} \leq \Theta_{i-1,h-1} \leq \Theta_{i-1,h} = U_{ih}. \quad (\text{A.6})$$

Thus Eq. (A.5) lies inside a non-empty interval and simultaneously satisfies $\Theta_{i,h-1} \leq \Theta_{ih} \leq \Theta_{i-1,h}$. Induction over $i + h$ therefore proves the coefficient inequalities in Eq. (7) for the complete grid. Finally, $D = \Theta - \Pi$ is computed deterministically.

For completeness, the derivative identity for a degree- d_x B-spline surface has the form

$$\frac{\partial g}{\partial x} = \sum_{i=1}^{K_x-1} \sum_{h=1}^{K_Z} \rho_{x,i} (\Theta_{i+1,h} - \Theta_{ih}) \tilde{B}_i(x) C_h(Z), \quad \rho_{x,i} > 0, \quad (\text{A.7})$$

where $\tilde{B}_i$ is a non-negative degree- $(d_x - 1)$ basis function. The corresponding Z derivative contains $\rho_{Z,h} (\Theta_{i,h+1} - \Theta_{ih}) B_i \tilde{C}_h$ with $\rho_{Z,h} > 0$. The coefficient orders therefore establish the derivative signs everywhere in the spline domain, including the appropriate one-sided limits at its boundaries.